Module 6 Week A — Stretch: Custom NER Rules

Honors Track. This is an Honors Track stretch assignment. Stretch assignments are for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. If you are behind on core work, focus there first.

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: None – this stretch builds on base lab work from Week A.

The Challenge

spaCy's built-in Named Entity Recognition (NER) model misses domain-specific entities because they were not in its training data. Terms like "COP28", "Paris Agreement", "IPCC AR6", and "2°C target" are meaningful climate concepts, but the base model either misclassifies them or ignores them entirely.

Build a custom spaCy `EntityRuler` that extends the base NER model with domain-specific entity patterns for climate terminology. Define patterns that capture entities the base model misses, integrate the EntityRuler into the spaCy pipeline, and measure the impact on entity extraction quality.

This is a real-world Natural Language Processing (NLP) engineering skill. Production NER systems almost always combine pre-trained models with domain-specific rules to handle terminology that general-purpose models were never trained on.

What You'll Learn

- Designing entity patterns with spaCy's `EntityRuler` (token-level and phrase-level)
- Custom entity labels beyond the standard NER schema (e.g., `CLIMATE_EVENT`, `POLICY`, `REPORT`, `THRESHOLD`)
- Pipeline composition: inserting a rule-based component before or after the statistical NER
- Measuring the precision/recall impact of custom rules on a real dataset
- The engineering tradeoff between rules and models in production NER

Outcome

Your deliverable extends your Lab 6A repo (`m6-l6a-{your-username}` , the GitHub Classroom repo from the lab — `lab-6a-ner-pipeline` is the lab's working branch, not the repo name) and adds:

Artifact	Description
Custom EntityRuler script	Defines at least 10 domain-specific entity patterns covering climate terminology
Before/after comparison	Entity counts and types extracted with and without the custom rules
Evaluation delta	Precision, recall, and F1 change on the gold-standard subset, **measured on overlapping standard-label entities only** (ORG, GPE, DATE, LAW, MONEY, PERSON, QUANTITY, LOC, EVENT, WORK_OF_ART) — see Constraints below
1-paragraph analysis	How the custom rules changed the entity landscape – where they helped, where they introduced noise

Constraints:

- Define at least **10 distinct EntityRuler pattern entries** across at least 3 custom entity types (e.g., `CLIMATE_EVENT` , `POLICY` , `REPORT` , `THRESHOLD` , `AGREEMENT`). Each entry passed to `ruler.add_patterns([...])` counts as one pattern — "Paris Agreement" defined once as a phrase pattern is one entry; the same string defined again as a token pattern is a second entry. The 10 entries must cover at least 8 distinct concepts (no more than 2 entries describing the same underlying entity)
- Patterns should be specific enough to avoid false positives – "Paris" alone should not trigger `AGREEMENT` ; "Paris Agreement" should
- Test the EntityRuler in two pipeline positions: before the built-in NER and after it. Document the difference in behavior
- Evaluate on the gold-standard subset (`gold_entities.csv`). The gold standard contains only standard spaCy labels (ORG, GPE, DATE, LAW, MONEY, PERSON, QUANTITY, LOC, EVENT, WORK_OF_ART), so report precision, recall, and F1 **on overlapping standard-label entities only** — a custom label like `CLIMATE_EVENT` cannot match a gold `EVENT` and including it will artificially depress precision. For your custom labels, the evaluation is **qualitative**: surface a few example texts where each custom rule fired, and discuss whether the match was correct
- The analysis must reference specific examples from the dataset – not generic observations

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Tips

- Start by running the base NER on 10-20 texts and manually noting what it misses. Use those missed entities to design your first patterns.
- spaCy's `EntityRuler` supports both token patterns (list of token attributes) and phrase patterns (exact string matches). Phrase patterns are simpler for named entities like "Paris Agreement"; token patterns are more flexible for variable-form entities like temperature thresholds.
- When the EntityRuler runs *before* the statistical NER, its matches take priority. When it runs *after*, the statistical NER's matches take priority. Which position gives better results depends on your pattern specificity.
- Custom rules can introduce false positives. Your analysis should honestly assess cases where rules fired incorrectly, not just count the improvements.

Submission

1. In your Lab 6A repo (`m6-l6a-{your-username}`), create a new branch named `stretch-custom-ner` off `main`
2. Add your EntityRuler script (e.g., `stretch_custom_ner.py`), the before/after comparison output, and your written analysis (e.g., `stretch_analysis.md`)
3. Push the branch and **open a separate pull request from `stretch-custom-ner` to `main`** — do not add these commits to your existing Lab 6A PR. The stretch PR can stay open; it will not affect the lab autograder
4. Paste the **stretch PR URL** (not the lab PR URL, not the bare repo URL) into TalentLMS -> Module 6 Week A -> Stretch 6A-S1